Tribhuvan University

Institute of Science and Technology

2078

✡

BScCSIT Level/First Semester
Mathematics[MTH 112]
Calculus

Full Marks: 80
Pass Marks: 32
Time 3 Hrs.

Candidates are required to give their answers in their own words as far as practicable.

Group A $(10 \times 3 = 30)$

Attempt any **THREE** questions.

1. (a) If $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$ then find fog and its domain and range. [5]

   (b) A rectangular storage container with an open top has a volume of $20m^3$. The length of its base is twice its width. Material for the base costs $Rs10$ per square meter; material for the sides costs $Rs4$ per square meter. Express the cost of materials as a function of the width of the base. [5]

2. (a) Using rectangles, estimate the area under the parabola $y = x^2$ from 0 to 1. [5]

   (b) A particle moves along a line so that its velocity v at time t is [5]

$$v = t^2 - t + 6$$

   (i) Find the displacement of the particle during the time period $1 \leq t \leq 4$.
   (ii) Find the distance traveled during this time period.

3. (a) Find the area of the region bounded by $y = x^2$ and $y = 2x - x^2$. [5]

   (b) Using trapezoidal rule, approximate $\int_1^2 1/x dx$ with $n = 5$. [5]

4. (a) Solve: $y' = x^2/y^2, y(0) = 2$. [5]

   (b) Solve the initial value problem: $y'' + y' - 6y = 0, y(0) = 1, y'(0) = 0$. [5]

Group B $(10 \times 5 = 50)$

Attempt any **TEN** questions.

5. Recent studies indicate that the average surface temperature of the earth has been rising steadily. Some scientists have modeled the temperature by the linear function $T = 0.03t + 8.50$, where T is temperature in degree centigrade and t represents years since 1900. (a) What do the slope and T-intercept represent? (b) Use the equation to predict the average global surface temperature in 2100.

6. Find the equation of tangent at $(1, 2)$ to the curve $y = 2x^2$.
7. State Rolle's theorem and verify the Rolle's theorem for $f(x) = x^2 - 3x + 2$ in $[0, 3]$.
8. Use Newton's method to find $\sqrt[6]{2}$ correct to five decimal places.
9. Find the derivative of $\mathbf{r}(t) = (1 - t^2)\mathbf{i} - te^{-t}\mathbf{j} + \sin 2t\mathbf{k}$ and find the unit tangent vector at $t = 0$.
10. Find the volume of the solid obtained by rotating about the y-axis the region between $y = x$ and $y = x^2$.
11. Solve: $y' + 2xy - 1 = 0$
12. What is a sequence? Is the sequence

$$a_n = \frac{n}{\sqrt{5+n}}$$

convergent?

13. Find a vector perpendicular to the plane that passes through the points: $P(1, 4, 6), Q(-2, 5, -1)$ and $R(1, 1, -1)$
14. Find the partial derivatives of $f(x, y) = x^2 + 2x^3y^2 - 3y^2 + x + y$, at $(1, 2)$.
15. Find the local maximum and minimum values, saddle points of $f(x, y) = x^4 + y^4 - 4xy + 1$.